DANAK
PROD Reg. nr 7026

FORCE Certification

EU Type Examination Certificate

No. 0200-NAWI-04955 Revision 3

W100 / W200 / W200IP64 / W200IP67 / WDOS / WDESK / WDESKLIGHT / WINOX / WTAB / W200BOX / W200BOXEC / ADPEW200 / WLIGHT

NON-AUTOMATIC WEIGHING INSTRUMENT

Issued by **FORCE Certification**
EU - Notified Body No. 0200

In accordance with the requirements in Directive 2014/31/EU of the European Parliament and Council.

Issued to **Laumas Elettronica SRL.**
Via Primo Maggio 6
I – 43022 Montechiarugolo (PR)
ITALY

In respect of Non-automatic weighing instrument designated W100 / W200 / WDOS / W200IP64/ W200IP67 / WDESK / WDESKLIGHT / WINOX / WTAB / W200BOX / W200BOXEC / ADPEW200 / WLIGHT with variants of modules of weight transmitters, load receptors, load cells and peripheral equipment.
Accuracy class III and IIII
Maximum capacity, Max: From 0.3 kg up to 999 999 kg
Verification scale interval: e = Max / n
Maximum number of verification scale intervals: $n \leq 10000$ for single-interval and $n \leq 3 \times 10000$ for multi-range and multi-interval (however, dependent on environment and the composition of the modules).
Variants of modules and conditions for the composition of the modules are set out in the annex.

The conformity with the essential requirements in annex 1 of the Directive is met by the application of OIML R76:2006, and EN 45501:2015.

Note: This certificate is a revised edition which replaces previous editions.

The principal characteristics and approval conditions are set out in the descriptive annex to this certificate.

The annex comprises 29 pages.

Jens Hovgård Jensen
2025-01-23

Digitally signed by Jens Hovgård Jensen
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Certification Manager

Issued on **2025-01-23**
Valid until **2031-03-08**

FORCE Certification references:
Task no.: 124-26512.90.40 and ID no.: 0200-NAWI-18773-1

Signatory: Jens Hovgård Jensen

Descriptive annex

Contents

1. Name and type of instrument and modules

The weighing instrument is designated W100 / W200 / W200IP64 / W200IP67 / WDOS / WDESK / WDESKLIGHT / WINOX / WTAB / W200BOX / W200BOXEC / ADPEW200 /WLIGHT. It is a system of modules consisting of an electronic weighing indicator, connected to a weight transmitter or to a separate load receptor and to peripheral equipment such as printers or other devices, as appropriate. The instrument is a Class III or IIII, self-indicating weighing instrument with single-interval, multi-range or multi-interval.

The WDOS indicator is also available in a price computing version intended to be used as a self-service instrument for direct sale to the public.

The name of the instruments may have the designation "JOLLY" put in front and may be followed by alphanumeric characters for technical, legally or commercial characterisation of the instrument.

The indicators consist of analogue to digital conversion circuitry, microprocessor control circuitry, power supply, keyboard, non-volatile memory for storage of calibration and setup data, and a weight display contained within a single enclosure.

The modules appear from Sections 3.1, 3.2 and 3.3; the principle of the composition of the modules is set out in Sections 6.1 and 10.

2. Description of the construction and function

2.1 Construction

2.1.1 Indicator

The electronic indicator consists of two electronic boards: A main board bearing the microcontroller, the ADC and all other components and a display board.
Alternately can the main board be equipped with a LED display and have no display board.
The main board's microcontroller and analogue section are identical for all models.

Display and indicators

The weighing indicators have one of the following types of display and indicators,

- LED display with LED indication for: NET, zero, stable, weight unit (kg, g) and weighing range (W1, W2, W3), and 6 digits with a height of 10 to 20 mm depending of model.
  Model WDOS has in addition to the primary LED display an auxiliary dot-matrix LCD display, which on the price computing version is used for display of unit price and price to pay.
- LCD display with indication for: NET, zero, stable, weight unit (kg, g, t) and weighing range (W1, W2, W3), and 6 digits, whose height depends of the model. (only WDESK, WDESKLIGHT, WINOX AND WTAB models).
- Dot matrix LCD display with indication for: NET, zero, stable and weighing range (W1, W2, W3), and display of weight and weight unit. (only WDESK, WDESKLIGHT, WINOX, WTAB models).

The enclosure is made of stainless steel (WINOX models) or of plastics intended for front panel or autonomous mounting. Screw terminals for connection of power, load cell and interface cables are placed on the rear of the panel models and inside the enclosure for the other models.
ADPEW200 is a W200 indicator mounted in an EX-proof metal housing and with external keys on the front.
On the front of the enclosure are 4 to 62 keys - depending of model - for operating the functions of the indicator.

All instrument calibration and metrological setup data are stored in the non-volatile memory.

The indicators are power supplied with 12 - 24 VDC, or 230 VAC.

2.1.2 Weight transmitters, load receptors, load cells, and load receptor supports

Set out in Section 3.3.

2.1.3 Interfaces and peripheral equipment

Set out in Section 4.

2.2 Functions

The weight indicating instruments are microcontroller based electronic weight indicators that require the external connection of strain gauge load cell(s). The weight information appears in the digital display located on the front panel and may be transmitted to peripheral equipment for recording, processing or display.

The primary functions provided are detailed below.

2.2.1 Display range

The weight indicators will display weight from –Max to Max (gross weight) within the limits of the display capacity.

2.2.2 Display test

A self-test routine is initiated at power up. The test routine turns on and off all of the display segments and indicators to verify that the display is fully functional.

2.2.3 Zero-setting

Pressing the "ZERO" key for 3 seconds causes a new zero reference to be established and ZERO annunciator to turn on indicating the display is at the centre of zero.

Semi-automatic zero-setting range: 4 % of Max.
Automatic zero-tracking range: 4 % of Max.
Initial zero-setting range: 20 % of Max.

Zero-setting is only possible when the load receptor is not in motion.

2.2.4 Zero-tracking

The indicators are equipped with a zero-tracking feature which operates over a range of 4% of Max and only when the indicator is at zero (gross or net) and there is no motion in the weight display.

2.2.5 Tare

2.2.5.1 Semi-automatic tare

The instrument models are provided with a semi-automatic subtractive tare feature activated using the "TARE" key.

2.2.5.2 Preset tare

The indicators have a preset tare function for manually insertion of a tare value. The function can operate simultaneously with a semi-automatic tare, if the preset tare is activated first.
The value of the preset tare can be displayed temporary upon operator's request.

2.2.6 Printing

A printer may be connected to the optional serial data port. The weight indicator will transmit the current to the printer when the "PRINT" key is pressed or one of the digital inputs is closed.

The printing will not take place if the load receptor is not stable, if the gross weight is less than zero, or if the weight exceeds Max.

The WTAB indicator may be equipped with a built-in printer.

2.2.7 Alibi memory

The indicators are equipped with an alibi memory; however, the alibi memory must be enabled in the configuration of the indicator in order to be used.

When enabled pressing the "Print" key or receiving a trigger command from interface will store the displayed weight in alibi memory, if it is a stable legal weight not already stored.

The alibi memory is a circularly used buffer, but an alarm message is displayed each time the first record is overwritten.

The alibi memory can be read from an external unit connected via one of the serial interfaces.

2.2.8 Weighing unstable samples (optional)

The indicator has a function for weighing unstable samples. It can only be used, if it is enabled in the configuration.

2.2.9 Real time clock

The instrument models have a real time clock giving the opportunity for printout with day and time information.

2.2.10 Price computing

Indicators with an LCD or dot-matrix LCD can work as a price computing weighing indicator with a price look-up table. Selection of product/price is performed via the digital input interface or via a serial interface.
Models WDOS (special version), WDESK, WINOX and WTAB.

2.2.11 Operator information messages

The weight indicator has a number of general and diagnostic messages which are described in detail in the user's guide.

2.2.12 Software version

The software version is displayed during the start-up of the indicator.

The version format is xx.yy.zz, where xx is the legal version no., while yy and zz are major and minor version numbers for changes and corrections not influencing the legal function of the software.

Software version for standard indicator is r1.yy.zz, where yy.zz must be 00.46 or higher.
Software version for indicator with price computing feature is r1.yy.zz, where yy.zz must be 03.04 or higher.

2.2.13 Event counter

The indicator has an event counter, which increment each time the calibration or legal part of the set-up has been changed. The value of the event counter can be displayed in the ‘Menu’ under ‘Info’.

2.2.14 Tilt limiting device

The indicator may be equipped with a tilt limiting device based on a digital input.

2.2.15 Gravity compensation

The gravity adjustment parameters can be used to compensate the weight difference between the place in which the instrument is calibrated and the place of usage.

2.2.16 Extended resolution

In a special service mode the indicator can display the weight with extended resolution (d = 0.1e).
The service mode is indicated with all the indication LED’s turned on.

3. Technical data

The W100/W200/WDOS/W200IP64/W200IP67/WDESK/ WDESK-LIGHT/WINOX/WTAB/W200BOX/W200BOXEC/ADPEW200/WLIGHT weighing instruments are composed of separate modules, which are set out as follows:

3.1 Indicator

The indicators have the following characteristics:

Type:	W100/W200/ W200IP64/W200IP67/WDOS/ WDESK/ WDESKLIGHT/WINOX/WTAB/ W200BOX/W200BOXEC/ADPEW200/ WLIGHT
Accuracy class:	III or IIII
Weighing range:	Single-interval, multi-range or multi-interval (2 or 3)
Maximum number of verification scale intervals (n):	10000 for Class III 1000 for Class IIII
Minimum input voltage per VSI:	0.2 µV
Maximum capacity of interval or range (Max_i):	$n_i \times e_i$
Verification scale interval, e_i = :	Max_i/n_i
Initial zero-setting range:	± 10 % of Max
Maximum tare effect:	100 % of Max
Fractional factor (pi):	0.5
Excitation voltage:	5 VDC
Circuit for remote sense:	Active, (see below)
Minimum input impedance:	43 ohm
Maximum input impedance:	1200 ohm
Connecting cable to load cell(s):	See Section 3.1.1
Supply voltage:	12 - 24 VDC, or 230 VAC
Operating temperature range:	Min/Max = -10 °C/+40 °C
Peripheral interface(s):	See Section 4

3.1.1 Connecting cable between the indicator and the junction box for load cell(s), if any

3.1.1.1 4-wire system

Line:	4 wires, shielded
Maximum length:	The certified length of the load cell cable, which shall be connected directly to the indicator.

3.1.1.2 6-wire system

Line: 6 wires, screened

Option 1:

Maximum length:	1315 m/mm^2 (for n = 10,000)
Maximum resistance per wire:	22.2 ohm

In case the (n) for the weighing instrument is less than (n) mentioned above, the following apply:

Option 2:

Coefficient of temperature of the span error of the indicator:	Es = 0.001 [%/25K]
Coefficient of resistance for the wires in the J-box cable:	Sx = 0.0015 [%/ohm]

L/A_{max} = 295.86 / Sx * (emp/n - Es) [m/mm²] in which emp = p'i * mpe * 100/e

From this, the maximum cable length for the weighing instrument may be calculated with regard to (n) for the actual configuration of the instrument.

Reference: See Section 10.

3.2 Indicator - EEx version

The indicators have the following characteristics when connected through shunt-diode safety barrier type MTL7761ac and MTL7766Pac:

Type:	W100/W200/ W200IP64/W200IP67/WDOS/ WDESK/ WDESK-LIGHT/WINOX/WTAB/W200BOX/ W200BOXEC/ADPEW200/WLIGHT
Accuracy class:	III or IIII
Weighing range:	Single-interval, dual-range or dual-interval
Maximum number of verification scale intervals (n):	7500 for Class III 1000 for Class IIII
Minimum input voltage per VSI:	0.52 µV
Maximum capacity of interval or range (Max_i):	$n_i \times e_i$
Verification scale interval, e_i = :	Max_i/n_i
Initial zero-setting range:	± 10 % of Max
Maximum tare effect:	100 % of Max
Fractional factor (pi):	0.5
Excitation voltage:	5 VDC
Circuit for remote sense:	Active, (see below)
Minimum input impedance:	43 ohm
Maximum input impedance:	1200 ohm
Connecting cable to load cell(s):	See Section 3.1.1
Supply voltage:	12 - 24 VDC, or 230 VAC
Operating temperature range:	Min/Max = -10 °C/+40 °C
Shunt-diode safety barrier:	MTL7761ac, MTL7766Pac, or similar
Peripheral interface(s)	See Section 4

3.2.1 Connecting cable between the indicator and the junction box for load cell(s), if any

3.2.1.1 4-wire system

Line:	4 wires, shielded
Maximum length:	The certified length of the load cell cable, which shall be connected directly to the indicator.

3.2.1.2 6-wire system

Line:	6 wires, screened

Option 1:

Maximum length:	1847 m/mm^2 (for n = 7,500)
Maximum resistance per wire:	31.2 ohm

In case the (n) for the weighing instrument is less than (n) mentioned above, the following apply:

Option 2:

Coefficient of temperature of the span error of the indicator:	Es = 0.002 [%/25K]
Coefficient of resistance for the wires in the J-box cable:	Sx = 0.0012 [%/ohm]

L/A_{max} = 295.86 / Sx * (emp/n - Es) [m/mm²] in which emp = p'i * mpe * 100/e

From this, the maximum cable length for the weighing instrument may be calculated with regard to (n) for the actual configuration of the instrument.

Reference: See section 10.

3.3 Weight transmitters, load receptors, load cells, and load receptor supports

Removable platforms shall be equipped with level indicators.

3.3.1 Weight transmitters

The W100 / W200 / W200IP64 / W200IP67 / WDOS / WDESK / WDESKLIGHT / WINOX / WTAB / W200BOX / W200BOXEC / ADPEW200 / WLIGHT indicators may be connected to a weight transmitter type TLM8 / CLM8 / TLB4 / TLB8 / TLK / TLKWF with an Evaluation Certificate according to WELMEC Guide 8.8:2011, EN45501:2015 and OIML R76:2006.

The indicator and the weight transmitter communicate as master-slave using standard Modbus RTU protocol.

3.3.2 General acceptance of modules

Any load cell(s) may be used for instruments under this certificate of type examination provided the following conditions are met:

1) There is a respective Part / Evaluation / Test Certificate (EN 45501) or an OIML Certificate of Conformity (R60:2000 or R60:2017) issued for the load cell by a Notified Body responsible for type examination under Directive 2014/31/EU

2) The certificate contains the load cell types and the necessary load cell data required for the manufacturer's declaration of compatibility of modules (WELMEC 2:2015), and any particular installation requirements). A load cell marked NH is allowed only if humidity testing to EN 45501 has been conducted on this load cell.

3) The compatibility of load cells and indicator is established by the manufacturer by means of the compatibility of modules form, contained in the above WELMEC 2 document, or the like, at the time of EU verification or declaration of EU conformity of type.

4) The load transmission must conform to one of the examples shown in the WELMEC 2.4 Guide for load cells.

3.3.3 Digital load cells

The instruments may be used with digital load cells via the RS485 interface. The following load cells are approved:

Manufacturer	Load cell designation	Cert. No.
Laumas Electronica S.r.L	COKD…, COD…	R60/2000-A-NL1-21.44
Laumas Electronica S.r.L	COLD	E-12.02.C09
Keli Sensing Technology (Ningbo) Co. Ltd.	ZSF-D, ZSF-DSS, ZSW-D, ZSW-DSS	R60/2000-NL1-13.25 Rev 1
TÉCNICAS DE ELECTRÓNICA Y AUTOMISMOS S.A.	740D	E-04.02.C06 Rev. 3

3.3.4 Platforms, weigh bridge platforms

Construction in brief:	All-steel or steel-reinforced concrete construction, surface or pit mounted
Reduction ratio:	1
Junction box:	Mounted in or on the platform
Load cells	Load cell according to Section 3.2.1
Drawings:	Various

3.3.5 Bin, tank, hopper and non-standard systems

Construction in brief:	Load cell assemblies each consisting of a load cell stand assembly to support one of the mounting feet bin, tank or hopper
Reduction ratio:	1
Junction box:	Mounted on dead structure
Load cell:	Load cell according to Section 3.2.1
Drawings:	Various

3.4 Composition of modules

In case of composition of modules, EN 45501:2015 annex F shall be satisfied.

3.5 Documents

The documents filed at FORCE (reference No. A530871, 118-30564 and 121-31619) are valid for the weighing instruments described here.

4. Interfaces and peripheral equipment

4.1 Interfaces

4.1.1 Load cell input

The indicators are as standard equipped with one load cell interface but can as an option be equipped with two load cell interfaces.

The connector pins for load cell connection are located on the rear of the enclosure (W100/W200/WDOS/WDESK/WINOX/WTAB/WLIGHT) or inside the enclosure (W200IP64/W200IP67/WDESK/WDESKLIGHT/WINOX/W200BOX/W200BOXEC/ADPEW200).

4.1.2 Other interfaces

The indicator may be equipped with one or more of the following protective interfaces,

- RS232C
- RS485
- Modbus RTU, Profibus, DeviceNet and CANopen Ethernet TCP/IP, Ethernet/IP, Modbus/TCP, Profinet IO, EtherCAT, POWERLINK, SERCOSIII, CC-link, CC-link IE and IO-link
- USB
- Digital output
- Digital input
- Analogue input
- Analogue output
- Wire-less

The interfaces are characterised “Protective interfaces” according to paragraph 8.4 in the Directive and do not have to be secured.

4.2 Peripheral equipment

Connection between the indicator and peripheral equipment is allowed by screened cable.

The instrument may be connected to any simple peripheral device with a CE mark of conformity.

5. Approval conditions

5.1 Measurement functions other than non-automatic functions

Measurement functions that will enable the use of the instrument as an automatic weighing instrument are not covered by this type approval.

5.2 Counting operation is not approved for NAWI

The count shown as result of the counting function is not covered by this NAWI approval.

5.3 Compatibility of modules

In case of composition of modules, EN 45501:2015 annex F shall be satisfied.

6. Special conditions for verification

6.1 Composition of modules

The environmental conditions should be taken into consideration by the composition of modules for a complete weighing instrument, for example instruments with load receptors placed outdoors and having no special protection against the weather.

The composition of modules shall agree with Section 5.3.

An example of a declaration of conformity document is shown in section 10.

7. Securing and location of seals and verification marks

7.1 Securing and sealing

Seals shall bear the verification mark of a notified body or alternative mark of the manufacturer according to ANNEX II, module F or D of Directive 2014/31/EU.

7.1.1 Indicator

Access to the configuration and calibration facility requires that a calibration jumper is installed on the main board, or that the operator types first a password and the key looked up on a special key card delivered by the manufacturer.
For the WLIGHT indicator a button on the rear of the indicator shall be pressed for access to the configuration and calibration facility.

The indicator has an event counter, which increment each time the configuration is changed.

Sealing of the cover of the enclosure - to prevent access to the calibration jumper and to secure the electronics against dismantling/adjustment - is accomplished with brittle plastic stickers. The sticker is placed so access to opening the enclosure is prohibited (see Figures 13, 14, 15, and 16).
The button on the WLIGHT indicator is covered by a plastic disc which is secured by a brittle plastic sticker (see fig. 17).

Alternatively, can the metal cover - covering the analogue circuits - on the mainboard be secured with a brittle plastic sticker (see Figures 18 and 19). This together with the fact that the serial number of the indicator is non-erasable stored on the mainboard for display on request, is sufficient to ensure the electronics against exchange.

7.1.2 Indicator - load cell connector - load receptor

Securing of the indicator, load receptor and load cell combined is done in one of the following ways:

- Inserting the serial number of the load receptor as part of the principal inscriptions contained on the indicator identification label
- The load receptor bears the serial number of the indicator on its data plate.

For W200IP64, W200IP67, WDESK, WDESKLIGHT, WINOX, W200BOX, W200BOXEC and ADPEW200 with connector pins for load cell connection inside the enclosure, this is not necessary, if the indicator enclosure is secured by sealing.

7.1.3 Junction box for load cells

A junction box for load cells shall be sealed against opening with wire and seal or brittle plastic sticker(s).

7.1.4 Peripheral interfaces

All peripheral interfaces are “protective”. Via the interfaces zero, span adjust and modification of the legally relevant parameters can be performed, similar to what the operator can, but it will increment the event counter as evidence for the infringement of the sealing. Apart from this, the interfaces neither allow manipulation with weighing data or legal setup, nor change of the performance of the weighing instrument in any way that would alter the legality of the weighing.

7.2 Verification marks

7.2.1 Printers used for legal transactions

Printers covered by this type examination certificate and other printers according to section 4.2, which have been subject to the conformity assessment procedure, shall not bear a separate metrological marking in order to be used for legal transactions.

8. Location of CE mark of conformity and inscriptions

8.1 Indicator

8.1.1 CE mark

CE mark and supplementary metrological marking shall be applied to the indicator according to article 16 of Directive 2014/31/EU.

8.1.2 Inscriptions

Manufacturer's trademark and/or name and the type designation is located on the front panel overlay.

Indelibly printed on a brittle plastic sticker located on the front panel overlay:

- Max, Min, e =

On the inscription plate:

- Manufacturer's name and/or logo, postal address, model no., serial no., type-approval certificate no., accuracy class, electrical data and other inscriptions.
- The value of the event counter (see Section 2.2.13).

Left to the manufacturer choice as provided in Section 7.1.2:

- Serial no. of the load receptor

8.1.2.1 Load receptors

On a data plate:

- Manufacturer's name, type, serial number, capacity

Left to the manufacturer choice as provided in Section 7.1.2:

- Serial no. of the indicator

9. Pictures

Figure 1 W100 indicator.

Figure 2 W200 indicator.

Figure 3 W200IP64 indicator.

Figure 4 W200IP67 indicator.

Figure 5a WDOS indicator.

Figure 5b Price computing version of WDOS

Figure 6a WDESK indicators

Figure 6b WDESKLIGHT indicator

Figure 7a WINOX indicators

Figure 7b WINOX desktop version indicator

Figure 8 WTAB indicator

Figure 9 W200BOX indicator.

Figure 10 W200BOXEC indicator.

Figure 11 ADPEW200 indicator

Figure 12 WLIGHT indicator

Figure 13 Sealing of W100 enclosure.

Figure 14 Sealing of W200 enclosure.

Figure 15 Sealing of W200IP64 and W200IP67 enclosures.

Figure 16 Sealing of WDOS.

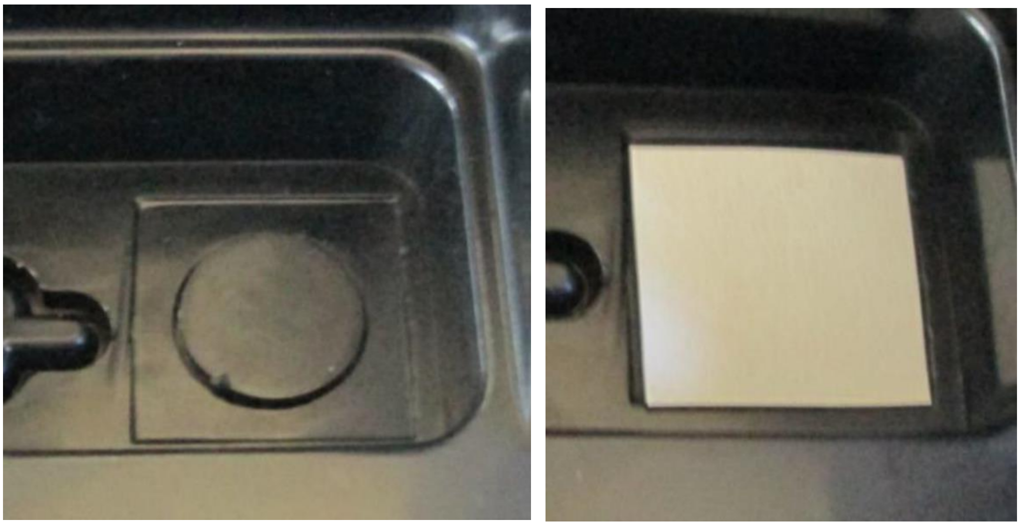

Figure 17 Sealing of WLIGHT.

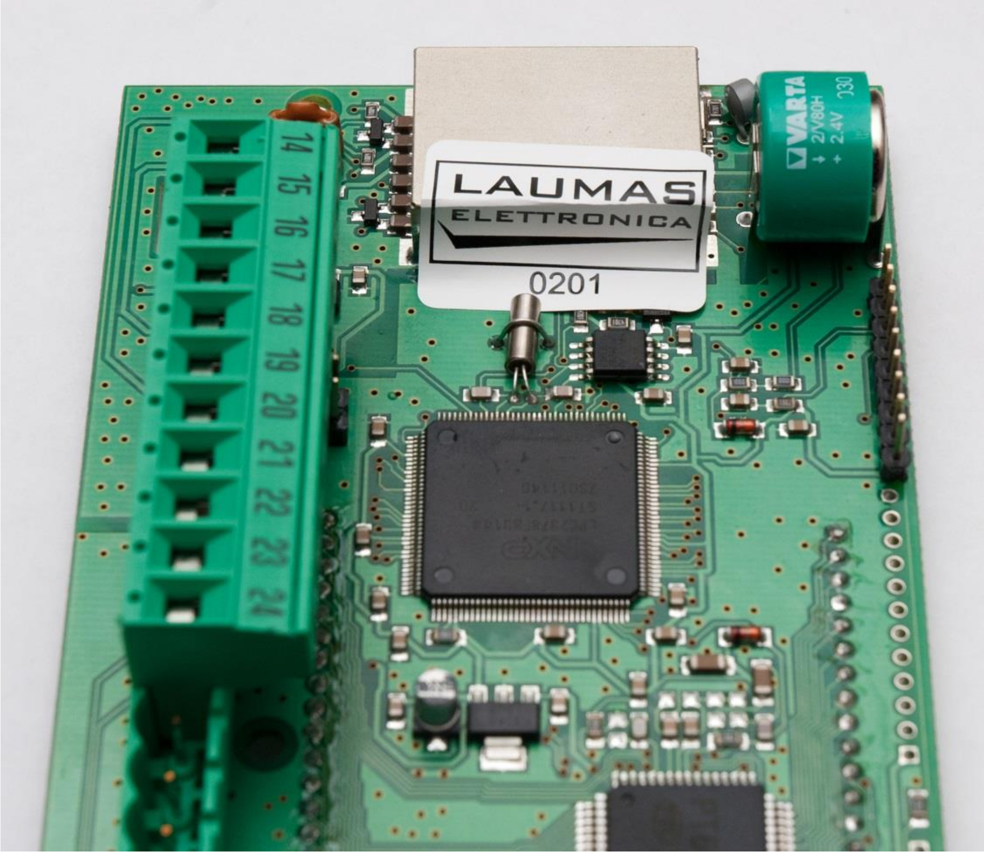

Figure 18 Sealing of Board in WDESK Light

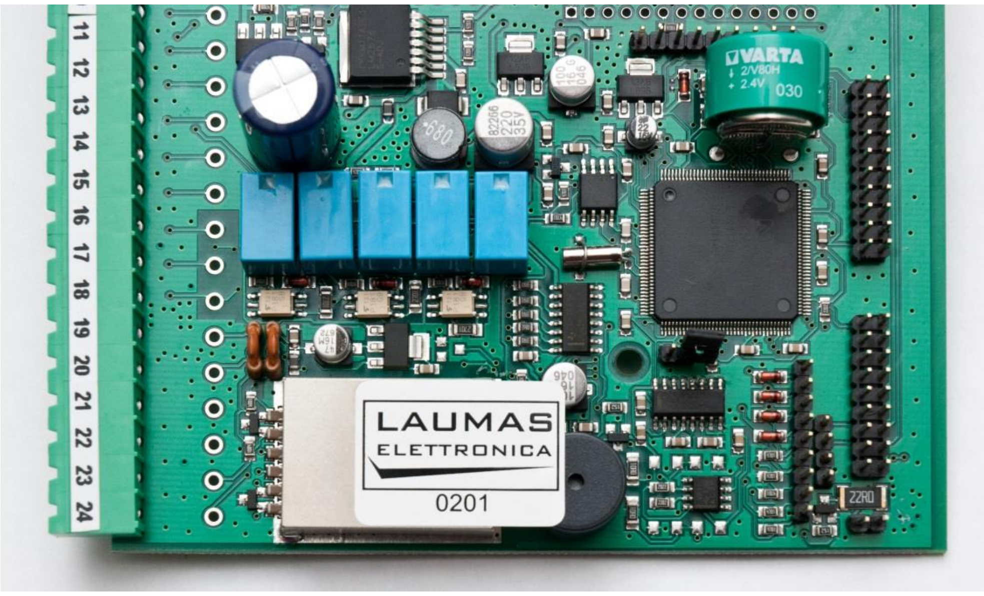

Figure 19 Sealing of board in WDESK, WINOX WTAB, W200BOX, W200BOXEC and ADPEW200

10. Composition of modules – an example

COMPATIBILITY OF MODULES

Ref.: WELMEC 2

Non-Automatic Weighing Instrument, multi-range.

Certificate of EU Type Examination N°: TEC: 0200-NAWI-04955

INDICATOR — A/D (Module 1) — Type: WDOS

Accuracy class according to EN 45501 and OIML R76:	Class$_{ind}$	(I, II, III or IIII)	III	
Maximum number of verification scale intervals (n_{max} or lower):	n_{ind}		10000	
Fraction of maximum permissible error (mpe):	p_1		0,5	
Load cell excitation voltage:	U_{exc}	[Vdc]	5	
Minimum input-voltage per verification scale interval:	Δu_{min}	[µV]	0,2	
Minimum load cell impedance:	R_{Lmin}	[Ω]	43	
Coefficient of temperature of the span error:	Es	[% / 25°C]		0,001
Coefficient of resistance for the wires in the J-box cable:	Sx	[% / Ω]		0,0015
Specific J-box cable-Length to the junction box for load cells:	$(L/A)_{max}$	[m / mm²]		1282
Load cell interface:	6-wire (remote sense)			
Additive tare, if available:	T^+	[% of Max]	0	
Initial zero setting range:	IZSR	[% of Max]	-10 / 10	
Temperature range:	T_{min} / T_{max}	[°C]	-10 / 40	
Test report (TR), Test Certificate (TC) or OIML Certificate of Conformity:				

LOAD RECEPTOR — (Module 2) — Type:

Construction:	Platform		
Fraction of mpe:	p_2		0,5
Number of load cells:	N		1
Reduction ratio of the load transmitting device:	R=F_M / F_L		1
Dead load of load receptor:	DL	[% of Max]	15
Non uniform distribution of the load: (NUD = 0 is acceptable)	NUD	[% of Max]	0
Correction factor:	$Q = 1 + (DL + T^+ + IZSR^+ + NUD) / 100$		1,25

LOAD CELL — ANALOG (Module 3) — Type: AZL

Accuracy class according to OIML R60:		Class$_{LC}$	(A, B, C or D)	C
Maximum number of load cell intervals:		n_{LC}		4000
Fraction of mpe:		p_3		0,7
Rated output (sensitivity):		C	[mV / V]	2
Input resistance of single load cell:		R_{LC}	[Ω]	409
Minimum load cell verification interval:	($v_{min\%}$ = 100 / Y)	$v_{min\%}$	[% of Emax]	0,005
Rated capacity:		E_{max}	[kg]	30
Minimum dead load, relative:		(E_{min} / E_{max}) * 100	[%]	1
Minimum dead load output return:	($DR_{\%}$ = 50 / Z)	$DR_{\%}$	[% of Emax]	0,0125
Temperature range:		T_{min} / T_{max}	[°C]	-10 / 40
Test report (TR) or Test Certificate (TC/OIML) as appropriate:				

COMPLETE WEIGHING INSTRUMENT — Multi-range

Manufacturer: — Type: WDOS with platform

Accuracy class according to EN 45501 and OIML R76:	Class$_{WI}$	(I, II, III or IIII)	III	
Fractions: $p_i = p_1^2 + p_2^2 + p_3^2$:	p_i		1,0	
Maximum capacity:	Max	[kg]	>>>>>>>>>>>>>>>>>	20
Maximum capacity for each partial weighing range:	Max_1 / Max_2	[kg]	8	20
Number of verification scale intervals for each weighing range:	n_1 / n_2		4000	4000
Verification scale interval for each weighing range:	e_1 / e_2	[kg]	0,002	0,005
Utilisation ratio of the load cell:	$\alpha = (Max_i / E_{max}) * (R / N)$		0,27	0,67
Input voltage (from the load cells):	$\Delta_u = C * U_{exc} * \alpha * 1000 / n$	[µV / e]	0,67	1,67
Cross-section of each wire in the J-box cable:	A	[mm²]	>>>>>>>>>>>>>>>>>	0,22
J-box cable-Length to the junction box for load cells:	L	[m]	>>>>>>>>>>>>>>>>>	4
Temperature range to be marked on the instrument: Not required	T_{min} / T_{max}	[°C]		
Peripheral Equipment subject to legal control:				

Acceptance criteria for compatibility				Passed, provided no result below is < 0		
Class$_{WI}$	<=	Class$_{ind}$ & Class$_{LC}$	(WELMEC 2: 1)	Class$_{WI}$:	PASSED	
p_i	<=	1	(R76: 3.5.4.1)	1 - p_i = >>>>>>>>>>>>>>>>>		0,0
n_i	<=	n_{max} for the class	(R76: 3.2)	n_{max} for the class - n_i =	6000	6000
n_i	<=	n_{ind}	(WELMEC 2: 4)	n_{ind} - n_i =	6000	6000
n_i	<=	n_{LC}	(R76: 4.12.2)	n_{LC} - n_i =	0	0
E_{min}	<=	DL * R / N	(WELMEC 2: 6d)	(DL * R / N) - E_{min} =	2,7	<<<<<<<<<<<<<<<<<
v_{min} * √N / R	<=	e_i	(R76: 4.12.3)	e_i - (v_{min} * √N / R) =	0,001	0,004
or (if v_{min} is not given)				Alternative solutions: ↑ ↓		
(E_{max} / n_{LC}) * (√N / R)	<=	e_i	(WELMEC 2: 7)	e_i - ((E_{max} / n_{LC}) * (√N/ R)) =		
Δu_{min}	<=	Δu	(WELMEC 2: 8)	Δu – Δu_{min} =	0,47	1,47
R_{Lmin}	<=	R_{LC} / N	(WELMEC 2: 9)	(R_{LC} / N) - R_{Lmin} = >>>>>>>>>>>>>>>>>		366
L / A	<=	$(L / A)_{max}^{WI}$	(WELMEC 2: 10)	$(L / A)_{max}^{WI}$ - (L / A) = >>>>>>>>		3483
T_{range}	<=	T_{max} - T_{min}	(R76: 3.9.2.2)	(T_{max} - T_{min}) - T_{range} =		20
Q * Max * R / N	<=	E_{max}	(R76: 4.12.1)	E_{max} - (Q * Max * R / N) = >>>>>>>>>>>>>>>>>		5,0
$DR_{\%}$	<=	125 * e_1 / Max	(WELMEC 2: 6c)	(125 * e_1 / Max) - $DR_{\%}$ =	0,0000	<<<<<<<<<<<<<<<<<
or (if DR% is not given)				Alternative solutions: ↑ ↓		
0.4 * Max / e_1	<=	n_{LC}	(WELMEC 2: 6c)	n_{LC} - (0.4 * Max / e_1) =		

Signature and date:

Conclusion . . . . . ***PASSED***

This is an authentic document made from the program: "Compatibility of NAWI-modules version 3.2".